SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING - CSA17 ARTIFICIAL INTELLIGENCE LAB

Day 1 - Program 9	Course Outcome: CO4	Language: Python 3

Question

Write the python to implement Travelling Salesman Problem

Aim

To find the shortest tour that visits every city exactly once and returns to the starting city.

Algorithm / Explanation

The brute-force method generates all (n-1)! permutations of the cities, computes the length of each tour and keeps the minimum.

It is exact but impractical for large n (factorial growth); heuristics such as nearest neighbour are used for big instances.

Program

```
# Travelling Salesman Problem - brute force over all permutations
from itertools import permutations

cities = ['A', 'B', 'C', 'D']
dist = {
    ('A', 'B'): 10, ('A', 'C'): 15, ('A', 'D'): 20,
    ('B', 'C'): 35, ('B', 'D'): 25, ('C', 'D'): 30,
}
def d(a, b):
    return dist.get((a, b)) or dist[(b, a)]

def tour_length(tour):
    return sum(d(tour[i], tour[i + 1]) for i in range(len(tour) - 1))

start = 'A'
best_tour, best_cost = None, float('inf')
for perm in permutations([c for c in cities if c != start]):
    tour = (start,) + perm + (start,)
    cost = tour_length(tour)
    print(" -> ".join(tour), "| cost =", cost)
    if cost < best_cost:
        best_tour, best_cost = tour, cost

print("\nShortest tour:", " -> ".join(best_tour))
print("Minimum cost :", best_cost)
```

Output

```
A -> B -> C -> D -> A | cost = 95
A -> B -> D -> C -> A | cost = 80
A -> C -> B -> D -> A | cost = 95
A -> C -> D -> B -> A | cost = 80
A -> D -> B -> C -> A | cost = 95
A -> D -> C -> B -> A | cost = 95

Shortest tour: A -> B -> D -> C -> A
Minimum cost : 80
```

Result

Thus the Python program was executed successfully and the output was verified.